DTU

DRILLING-TYPE UNDERREAMER

ENLARGEMENT

GENERAL DESCRIPTION

DRILLING-TYPE UNDERREAMER THAT ENLARGES PREVIOUSLY DRILLED PILOT HOLES. A BOTTOM BOX CONNECTION LETS A BIT OR A BULLNOSE RUN BELOW THE TOOL, SO THE DTU CAN DRILL AND UNDERREAM AT THE SAME TIME. MUD FLOW CAN BE SPLIT BETWEEN THE CUTTER ARMS AND THE BIT OR BULLNOSE. IT RUNS WITH WATER, MUD, AIR, AERATED MUD OR ANY OTHER CIRCULATING MEDIUM.

TECHNICAL CHARACTERISTICS

Tool series	Std. opening Ø	Optional opening Ø	Body Ø	Collapsed Ø	Through casing Ø x lb/ft	Fishing neck length	Fishing neck Ø	Overall length	Top pin API Reg.	Wt (lb)
3600	6	5-1/2 - 6	3-5/8	3-5/8	4-1/2 x 15	8	3-3/8	35	2-3/8	170
5700	8-3/4	7 - 8-3/4	5-3/4	6	7 x 38	18	4-3/4	70	3-1/2	360
7200*	11	9 - 11	7-1/4	7-1/2	8-5/8 x 40	18	5-3/4	74	4-1/2	770
8200*	14	10 - 14	8-1/4	8-1/4	9-5/8 x 53	18	5-3/4 or 8	79	4-1/2 or 6-5/8	900
9500*	15	12 - 15	9-1/2	10-1/4	11-3/4 x 71	18	8	82	6-5/8	1,150
11700*	17-1/2	14-3/4 - 20	11-3/4	11-3/4	13-3/8 x 92	20	8	96	6-5/8	1,670
15000*	17-1/2 - 26	17-1/2 - 26	14-3/4	14-3/4	16 x 75	20	8 or 9	97	6-5/8 or 7-5/8	2,800
17000*	32	24 - 32	17	17	18-5/8 x 78	20	9 or 10	87	7-5/8 or 8-5/8	3,000
22000	36	28 - 36	22	22	24-1/2 x 113	20	9 or 10	100	7-5/8 or 8-5/8	4,400

* Available with PDC Bearclaw cutters. Dimensions in inches, weights in lb (approximate). Optional opening diameters on request.

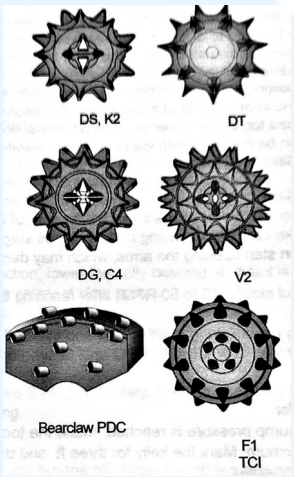
DESIGN FEATURES

- **Drill and underream in one run:** bottom box connection for a bit or bullnose.
- **Flexible hydraulics:** flow can be divided between the arms and the bit or bullnose; orifice jets are selected to suit the available hydraulics.
- **Cooling and cleaning:** jet placement keeps the cutters cool and aids annular lift of cuttings.
- **Extra flow options:** jetted bullnose and jetted top sub to divert more flow when needed.
- **Cutting structures:** open-bearing milled tooth (standard), TCI or PDC Bearclaw.

PERFORMANCE PARAMETERS

Rotary speed, reaching shoulder depth	30 - 50 rpm
Rotary speed, cutting the shoulder	80 - 150 rpm max.
Rotary speed, underreaming	80 rpm
Weight on tool	1,000 lb per inch of body Ø
Stabilization	Full-gauge stabilizer 60 - 90 ft above and below
Position in the BHA	Middle of the drill collars

CUTTER OPTIONS



Gauge diameter tolerances, arms open

Size (in)	Tolerance
6 - 9	+1/16 - 0
9-1/2 - 13	+3/32 - 0
14 - 18	+1/8 - 0
20 - 30	+3/16 - 0

Applies to arm sets of RTU, DTU, drag and XTU underreamers, all cutting structures.

HOW TO ORDER

Please specify: 1. tool series 2. top and bottom connections 3. fishing neck Ø 4. opening Ø 5. size and weight of casing to run through, if known 6. bullnose or bits (ordered separately) 7. cutting structure (milled tooth, TCI or PDC).